EPIC TECHNICAL SPECIFICATION BLUEPRINT

Adversarial Test Vector & Sovereign Governance

Framework Model:	EPIC Portfolio Specification Architecture
Core Runtime:	GraphBit DAG Executor & PyO3 Rust Validation Layer
Target Hardware:	NVIDIA Jetson Edge (8GB Shared RAM)
State Storage:	PostgreSQL Context Checkpointer

EVENT (THE BUSINESS RISK)

Enterprise SaaS platforms face acute compliance and operational vulnerabilities when deploying open source models at the edge. Native safety alignments are brittle. They are easily bypassed or altered via downstream quantization or weight modifications. Relying on a model to self govern its safety parameters exposes the corporate ecosystem to prompt injections, excessive agency, data leaks, and unpredictable blast radiuses. This fragility creates a critical barrier to enterprise adoption. It directly threatens Net Retention Revenue due to systemic security liabilities.

PERFORMANCE (THE STRATEGIC ACTION)

To validate the structural resistance of the local architecture, an Adversarial Evaluation Harness was engineered within the Phase 5 testing framework. Instead of using a standard aligned model, a pre-abliterated local variant that has had its internal refusal mechanisms surgically removed was deployed. The framework routes this uninhibited model through an isolated GraphBit Directed Acyclic Graph, subjecting the runtime to high risk, non-compliant validation datasets.

The defense matrix operates via three synchronized layers:

- **The In-Flight Sovereign Gate:** Processes raw token outputs through PyO3 Rust validation layers to catch toxic or unaligned payloads before they write to the application layer.
- **The Stigmergic Blacklisting Layer:** Drops negative vector trails directly into the PostgreSQL context checkpointer when contradictory logic loops occur.
- **The Deterministic Human Fallback:** Monitored via a global state machine, forcing an immediate state lock and human Centaur intervention the moment the system Frustration Index breaches the 75 percent threshold.

IMPACT (THE QUANTIFIED RESULT)

This architecture mathematically proves that platform safety can be achieved independently of model alignment. During adversarial simulation, the framework achieves a 100 percent mitigation rate against action risks and excessive agency. By executing this testing vector locally on the 8GB NVIDIA Jetson

Orin Nano, cloud egress costs for continuous safety evaluation are reduced to zero. Most importantly, the PyO3 Rust telemetry loop logs every failure vector with sub-1ms latency, passing 100 percent of the forensic audit trail to the human operator. This completely eliminates state drift and context repetition during escalations, protecting enterprise account health scores.

CONCLUSION (THE STRATEGIC TAKEAWAY)

True enterprise AI governance is forced at the orchestration layer, not requested at the model level. By utilizing an abliterated model as a dedicated testing adversary, the Centaur framework demonstrates to engineering executives that automated efficiency can coexist with absolute compliance. We do not trust the model to behave. We trust the system of structural resistance wrapped around it.

> *"We do not protect the platform by hoping the model behaves. We protect it by building architectural systems of structural resistance that force deterministic outcomes."*